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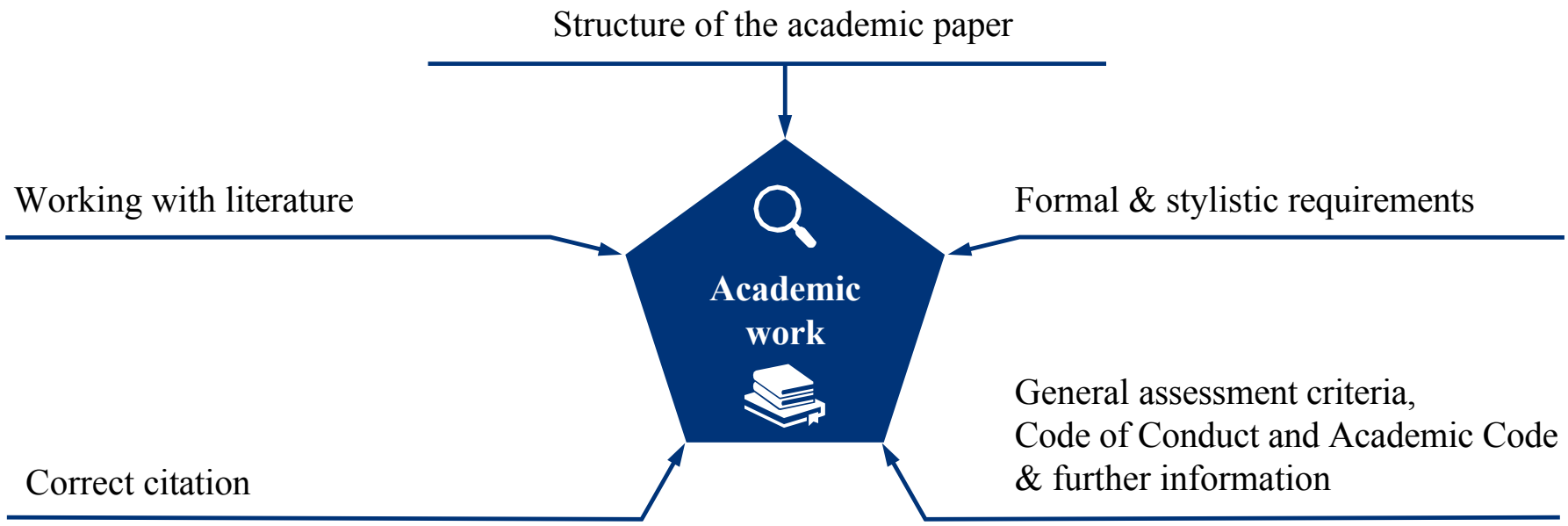
Seminars & dissertations

A guide to writing academic papers at the Chair of Business Administration, with a focus on digital transformation and start-ups

General | Introduction

The following guidelines are intended to serve as a guide for your academic work and to highlight potential sources of error.



In particular, the structure and content of the individual sub-sections should be adapted to suit the topic and scope of the work and agreed with the relevant supervisor.

Agenda

1 Structure of the academic thesis

2 Working with the literature

3 Formal and stylistic requirements

4 Correct citation

5 General assessment criteria, code of conduct and academic integrity & further information

1. Structure of the academic paper

Depending on the focus of the thesis (theoretical, empirical, systematic literature review), different structures may be appropriate. In general, however, you can use the following outline as a guide:

Table of contents

Abstract… I
Table of contents… II
List of Figures… III
List of tables… IV
List of abbreviations… V
1. Section 1 (Introduction) 1
1.1. Section 2 2
1.1.1. Section 3: Level 3 5
…
2. Outline point 1st level 10
…
Appendix (optional, but best avoided) 16
Bibliography… 20

→The second or third level of your outline should consist of at least two sub-sections (1.1., 1.2.,…)

→Your lists (with the exception of your bibliography) and the abstract must be numbered using Roman numerals.

→From the first section heading (Introduction) onwards, decimal numbers (starting with ‘1’)

1. Structure of the academic paper

It is advisable to include the following sections:

1. Structure of the research paper

The abstract is usually found before the introduction and the lists; it forms the "essence" of your work.

- Approx. 150–200 words; the text should be concise, precise and objective
- Do not use quotations
- Rule of thumb: one sentence per chapter
- Try to answer the following questions:
  - What is the aim?
  - What are the key findings?
  - Who will find this interesting?
  - How did they go about it?
- This list is intended to provide a general guide. Select the points that are relevant to your work and necessary for the reader to grasp its essence.
- Rather than listing all the implications, provide a concise conclusion
- Pay attention to grammatical tense: past tense for methods and results, present tense for conclusions and background

1. Structure of the academic paper

Generally serves as a brief and concise introduction to the topic, comprising between 5 and 10% of the total length of the thesis

- 1st paragraph: Introduce the reader to the topic of the paper: why is it interesting/relevant?
- 2nd paragraph: Identify gaps in the research: why does the topic addressed fill a gap in the research?
- 3rd paragraph: Aim of the paper
- Tip: Use the C.A.R.S. model by Swales (https://researchwrit.files.wordpress.com/2015/01/article_swales_cars-model.pdf)
- **Note:**
  - Don't give everything away; the reader should simply be made curious
  - Save definitions for the theory section
  - Do not promise too much and do not exaggerate

1. Structure of the academic paper

Empirical study:

- The aim of the theoretical section is to formulate the hypotheses: the entire text should be focused on this, i.e. everything should lead up to the hypotheses.
- The theoretical section forms the basis of the empirical work; therefore, a clear connection should be established between the theory and the empirical work.
- Hypotheses at the end of each section
- The reader is not interested in everything they have read, but only in the relevant aspects of your argument.
- You should explain why you have chosen a particular approach. Why were the variables used chosen, etc.?
- **Note:**
  - Do not skip any steps in your argument: A implies B implies C … Not: A implies C
  - Often too much literature review and too little derivation of the hypotheses
  - Use academic articles from highly ranked journals as a guide (A+/A, VHB Jourqual: http://vhbonline.org/vhb4you/jourqual/vhb-jourqual-3/)

1. Structure of the academic paper

Literature Review:

- Here, you do not need to formulate hypotheses, but rather prepare the reader thematically for your study
- Use only the most essential theories that the reader needs to understand your subsequent discussion.
- The reader is not interested in everything you have read, but only in the relevant aspects their reasoning.
- You should explain why you have chosen a particular approach.

1. Structure of the academic paper

Empirical study:

- Possible sections: Sample, development of research materials and pre-tests, research procedure, research plan, design and manipulation, instruments and procedures, dependent variable, manipulation check
- Instruments or procedures used (e.g. scales)
- For standard procedures (e.g.)
  - What are they intended to measure?
  - Number of items
  - Source (note: use original sources)
  - Sample items (not all items need to be listed here; include the complete scale in the appendix)
- For rare procedures, modified scales or self-developed scales: list all items and, where applicable, describe any adjustments in detail

1. Structure of the academic paper

Literature Review:

- Identification of relevant studies on the selected topic
- Description of the procedure
  - Search terms used in the search
  - Inclusion criteria for the selected studies
  - Time frame of the selected studies
  - Selection of publications
  - Selection of study quality
- Is the method used to select the literature reviewed described in a reproducible manner?

1. Structure of the scientific paper

Empirical study:

- In particular, the results necessary to answer the research question and hypotheses must be presented (not every possible correlation needs to be reported statistically)
- Negative results that do not support the hypotheses should also be documented and discussed later
- Often, a brief comment is made here as to whether the results confirm the researcher's own hypotheses or not, a No further interpretation is provided at this stage.
- Graphs and tables are useful here
- If reporting is purely descriptive, do not use the word 'significant'. Never write: 'this proves that…'
- Present descriptive results before presenting inferential statistics
- Always include the standard deviation with mean values
- Tables are useful for presenting descriptive statistics; for a small number of groups, this can also be done in text form (e.g. Training group X (M = 2.34, SD = .56) showed lower values than the control group (M = 3.38, SD = .35).

1. Structure of the research paper

Empirical study:

- Typical presentation: Test statistic (degrees of freedom) = result, significance (plus effect size)
  - $t(34) = 2.57, p < .05, d = 0.68$
  - $F(1, 34) = 6.60, p < .05, \eta^2 = .21$
  - $\chi^2(4) = 10.51, p < .05$
- Statistical abbreviations using Latin letters (F, t, M, SD, r, d, N, n, R^2) are written in italics
- Statistical abbreviations using Greek letters (α, β, η^2, χ^2) are not written in italics
- Entering two decimal numbers
- Insert a space between different components: $t(34) = 2.57$ instead of $t(34)=2.57$
- Tables and figures must also be labelled in accordance with APA guidelines
- **Please note:**
  - Never copy SPSS outputs and paste them as a graph or table; always create your own tables and figures
  - Do not use significance levels to 'prove' the magnitude of an effect

1. Structure of the scientific paper

Literature Review:

- Try to organise the works according to their content
- *Historical organisation:* here, the existing literature is presented chronologically. In this way, you can trace the development of a discourse or a specific subject area.
- *Conceptual/thematic organisation:* groups sources with the same or similar theoretical concepts or similar focal points.
- *Methodological organisation*: here, you consider works with comparable methodological approaches or procedures together.

1. Structure of the academic paper

- Structuring from specific to general
- First, summarise the results again in a single paragraph
- Then comes the most important paragraph: **(!)**
  - Linking the findings (in empirical studies: to the hypotheses) and comparing them with the study's objectives
  - Placing the results in a broader context
- The aim of the discussion is to situate the results within the current state of research, as previously outlined (Introduction & Theoretical Background).
- One begins with the specific results of the study and then moves on to placing the results within a broader context.

1. Structure of the academic paper

- The implications can also be included as a sub-section of the discussion.
- Explain why your work and its findings are relevant.
- Here, a distinction can be made between added value (= ‘Contribution’) for academia (‘Theoretical Implications’) and for practice (‘Managerial Implications’).
- Rule of thumb: try to identify at least three ‘theoretical’ and three ‘managerial implications’ respectively.

1. Structure of the academic paper

- This section discusses potential weaknesses of the study: however, do not be overly critical and Please only mention weaknesses in the work that you could not have avoided during your research
  .
- Formulate interpretations, criticism, etc. cautiously (‘hedged conclusion’)
  “This might be an effect of…”, “Another reason for that observation may be…”

- A good way to deal with limitations is to identify new research opportunities.
  - What questions may still remain unanswered?
  - How will scientific research on your topic proceed from here?
  - Do not make this paragraph too visionary; instead, try to consider/describe/outline what future developments might look like, based on your findings.

2. Working with literature

Strategies for literature research

Systematic method	If a relevant specialist article is found in a journal, a systematic review of older volumes of the journal is carried out, or one looks to see what renowned authors have published on the respective topic.
Method of concentric circles	Search based on a known source (Note: ensure the original source is up to date, otherwise important current sources may be overlooked)

- Combining both strategies makes sense!
- Make sure you document your strategy in detail so that, in theory, anyone can understand how you came across the articles.

2. Working with literature

Citable

- Sources must generally be public and accessible
- Search for primary sources
- The following may not be cited: lecture notes, seminar papers and examination papers by students, verbal information and unpublished, internal documents
- BUT not all generally publicly accessible sources are citable
  - No popular science literature
  - Practical guides ("10 Steps to Success", "The ABCs of…")
  - Textbooks only in exceptional cases
  - Only cite online sources if printed sources are not available

    !

Not citable

- The following details or sources of information are generally not suitable for citation and should therefore not be used in your paper:
  - e.g. Wikipedia or other wikis
  - Lecture notes
  - Other assignments

3. Formal and stylistic requirements

It is recommended that you use different tenses for specific sections of the work.

Past

- What other authors have found or said (*"X and Y were able to show that…", "over the last 10 years, numerous models have been developed…"*)
- Method section (Exception! See Present) (*"A a test involving 200 participants…"*)

Present

- Theory and findings (*"The term X refers to…", "these theories share the assumption that…"*)
- Discussion of one's own results (*"On examining Table X, a negative correlation becomes apparent…"*)
- Conclusions
- In the methods section: when describing established testing procedures established test procedures

Future

- Where applicable, "future research"

3. Formal and stylistic requirements

Form

- Font: Times New Roman (Ensure consistent use of the font, including in figures!)
- Font size 12 / Footnotes (in font size 10 if necessary)
- Line spacing 1.5
- Justified
- Page margins 2.5 cm (top, bottom, right, left)

Style

- Clear, unambiguous and precise wording
- Sentences should not be longer than three lines
- Avoid unnecessary filler words
- Avoid colloquial expressions
  "always and forever", "somehow", "of course", "incredible", "enormous", "solely and exclusively", "never ever", "of course", "I think…", "I suppose…"…
- Avoid everyday and colloquial language
- No exaggerations (e.g. not "incredibly high costs")
- Not too many unnecessary foreign words
- No use of the first-person singular or plural
- Avoid using the word "one"
- Choose and position headings sensibly (not at the bottom of a page!)

3. Formal and stylistic requirements

Graphics

- Present the key messages of graphics and tables in the text
- Provide graphics and tables with meaningful captions
- Always cite your sources
- Label adapted material with the note “based on”
- Captions should be marked with the abbreviation “Fig.” or “Table” and numbered consecutively (“Fig. 1: Title”)
- The titles are included in the list of figures and the list of tables
- If the figure is cited from another source, a source reference in the form of a short citation must be included directly below the figure or table and after the title (in brackets)
- The sources of the figures are listed in the bibliography and NOT in a footnote
- Sometimes it is necessary to provide the figure with a caption; in this case, it is placed below the figure but before the source reference

4. The correct citation

Excerpt from the APA citation guidelines for academic papers (6th Edition/2010: https://apastyle.apa.org/manual/)

General:

- The following rules are based on the guidelines of the American Psychological Association (APA)
- These apply to all natural and social science disciplines, i.e. to all international publications and journals in this field
- Consistent and standardised citation!
- Both verbatim or direct quotations (passages reproduced word for word) and indirect quotations (the adoption of an idea) must be clearly identified!
- Every source used in the work (book, contribution to an anthology, journal article, electronic source) must be cited!
- A brief reference in the text must be followed by a full citation in the bibliography
- The quotation/reference must be verifiable and clearly traceable for the reader
- Use reference management software such as EndNote, Citavi, Mendeley or Zotero.

4. Correct citation

Indirect quotation

- An author is often quoted only in essence, not verbatim
  - Indirect quotations are **not** placed within quotation marks
  - 'cf.' is not used (except, for example, when referring to several studies, etc.)
- The source reference is always given before the punctuation mark
  Example: … is a specific procedure (Bartlett, 1932, p. 14).
- Authors with the same surname:
  - Add the initials of the first name
    S. T. Smith (2000) and J. D. Smith (1999)
    - Two or more works within a single entry:
  - In alphabetical order as they appear in the reference list, separated by a semicolon
    (Jones, 2003; Thomas, 2010)
  - If by the same author, then by year (Jones, 2003, 2007)

4. The correct citation

Indirect quotation

- Three types of short receipt:
  - **Work by a single author**

    1. … an early examination of this phenomenon (Bartlett, 1932) …
    2. … Bartlett (1932) had already examined this phenomenon …
    3. As early as 1932, Bartlett was studying this particular phenomenon …
  - **If a text has two authors** --> cite both *Example:* (Blondin & Waller, 2006)
  - **For 3 to 5 authors** --> cite all of them the first time, then only mention the first author and add the abbreviation et al.

    Example: (Baschek, Bredenkamp, Öhrle & Wippich, 2001) then (Baschek et al., 2001)
  - **If there are more than 5 authors** --> generally cite only the first author *Example:* (Baschek et al., 2001)
  - **For multiple sources** --> separate the different sources with a semicolon and list them in alphabetical order

    Example: Several studies (Murray, 1970; Smith et al., 1990) show that …

4. Correct citation

The direct quotation

- Direct quotations must be reproduced word for word and placed within quotation marks!
- Example of a **direct quotation:**
  Higher education is the ‘sum of expanded material and intellectual life opportunities’ (Wehler, 1980, p. 127).
- **Important:** It is essential to include the page number!
- Quote verbatim passages of more than 40 words in a separate paragraph without quotation marks, indent the text
  indented -->
  Example of a **block quotation:**
  Traxel (1974) provides the following description:
  Modern psychology sees itself as an empirical science. This statement applies insofar as … process it differently. (p. 15)*

* Except for block quotations, the source reference for direct quotations is given before the punctuation mark!

Secondary quotation

- Secondary literature should be the exception!
  - Cite sources -->
    Example of a **secondary citation:**
    In a study by Piaget (1968, cited in Berk, 2005, p. 76) …
    → **Berk** is listed in full in the bibliography, Piaget is not

4. Correct citation

Entries in the bibliography

- List all materials used
  - **arranged alphabetically** by author's surname (within an author's works, chronologically, oldest source first)
  - Do not distinguish by source type
  - Formatting without bullet points (hanging indent!)
- Information must be **accurate** and **complete**!
- Each reference usually includes: author(s), year of publication, title, publication details

Journals

Author, A., Author, B. & Author, C. (Year). Title of the article. **Title of the journal**, Volume, page number.

Reichle, B. & Gloger-Tippelt, G. (2007). Family contexts and social-emotional development. *Childhood and Development*, 16 (4), 199–208.

16 = Volume number not in brackets / (4) = Issue number in brackets, if available DOIs do not need to be included and may be uniformly omitted

Theses

Köster, J. (2010). *Effective journalistic quality management?* Unpublished dissertation, Technical University of Ilmenau.

4. Correct citation

Books

- Author, A. (Year). *Title of the work.* Place: Publisher.

  Bourdieu, P. & Passeron, J.-C. (1971). *The Illusion of Equal Opportunity.* Stuttgart: Klett.

- **Additional information** about the title, such as '2nd edition' or 'Volume 3' --> **after the title** (not in italics, as it is not part of the title).

  Brockett, O. (1987). *History of the Theatre* (5th ed.). Boston: Allyn and Bacon.

- **Edited by:**

  Berg, T. (ed.). (2002). *Modern Election Campaigns: A Look Behind the Scenes.* Opladen: Leske + Budrich.

- **Book chapter or contribution:**
  - Author, A. (Year). Title of chapter. In B. Author (ed.), *Title of work* (page number). Place: Publisher.

  Sander, E. (1997). The stereotype of the poor pupil: A review of the literature. In F. E. Weinert & A. Helmke (Eds.), *Development in primary school age* (pp. 261–271). Göttingen: Hogrefe.

Website content

State Institute for Teacher Training and School Development (2011). *Family Literacy (FLY).* Available at http://li.hamburg.de/family-literacy [08.04.2013].

Federal Agency for Civic Education (2010). *Elections in Germany.* Available at http://www.bpb.de/wissen/7004AT,0,Wahlen_in_Deutschland.html [08.04.2013].

http://www.bpb.de/wissen/7004AT,0,Wahlen_in_Deutschland.html [8 April 2013].

5. General assessment criteria

Assessment criteria

- Description of the topic and the aim of the work
- Logical structure and structure of the paper
- Treatment of the topic, coherent argumentation, theoretical foundation, etc.
- References to further contexts and personal reflections
- Formal presentation (typos, etc.)
- Language and spelling
- Selection of and engagement with literature
- Linking theory and practice

5. Code of conduct and academic integrity

Academically correct conduct when writing of academic papers at the Chair of Business Administration, in particular Technology and Innovation Management (TIM) – EXCERPT

Misrepresentation of any kind	▪ Falsification of data ▪ Incorrect information in a job application or a grant application ▪ Disposal of primary data, insofar as this contravenes statutory provisions or recognised disciplinary principles of scientific work are violated
Infringement of the intellectual property rights of others	▪ A copyright-protected work created by another person, or significant scientific findings, hypotheses, etc.: unauthorised use whilst claiming authorship (plagiarism) ▪ Exploitation of research approaches and ideas, particularly as a reviewer/supervisor (theft of ideas) ▪ Falsification of content
Consequences	▪ **Failure of the academic work!**

Based closely on: Guidelines on the Prevention and Handling of Research Misconduct at Saarland University

5. Further information

If you still have questions, you may also find it helpful to consult the literature on academic work:

- Bänsch, A., & Alewell, D. (2009). Scientific Writing. Oldenbourg Verlag, Munich.
- Gopen, G., & Swan, J. (1990). The Science of Scientific Writing: If the reader is to grasp what the writer means, the writer must understand what the reader needs. American Scientist, 78(6), 550–558.
- Ebster, C. & Stalzer, L. (2008). Academic Writing for Economists and Social Scientists. 3rd ed., Universitätsverlag, Vienna.
- Plümper, T. (2008). Efficient Writing: A Guide to Writing Qualification Papers and Academic Texts (2nd ed.). Munich, Germany: Oldenbourg Wissenschaftsverlag GmbH.
- Theisen, M. R. (2008). Academic Writing: Technique – Methodology – Form. 14th ed., Vahlen, Munich.
- Töpfer, A. (2010). Successful Research. A Guide for Bachelor's, Master's and PhD Students. 2nd ed., Springer, Heidelberg.